

POWER INTEGRITY - Key To Reliable And Optimised System Design

With modern-day complex multi-chipset systems, power delivery designs are becoming challenging and more critical than ever. Power integrity analysis helps us understand the impact of each component and its parasitic value on the final performance of the CPU and helps in designing an optimised system for reliable operation

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Power distribution of a system plays a crucial role in its reliability and performance. Be it any system ranging from mobile phones to laptops to servers to national grids that are powering up the nation, key to reliable system operation is to deliver right amount of current to its different loads at right voltage levels.

Imagine 0.7V DC voltage at CPU (central processing unit) power input that is rated for 1-1.5V DC for stable operation or 350V AC at air-conditioner input that is rated for 230V AC. It is a known fact what happens next. Either the CPU hangs due to low voltage or the air-conditioner gets damaged due to high voltage at input.

The question that arises next is, how to ensure electrical reliability as a system designer? Power integrity is the answer.

Power integrity ensures required voltage and currents are delivered from source to load within the system. Through power integrity analysis, one can understand voltage levels at various system loads under worst-case operating conditions, which is the ultimate test of electrical reliability. This article covers the power integrity analysis of electronic products only.

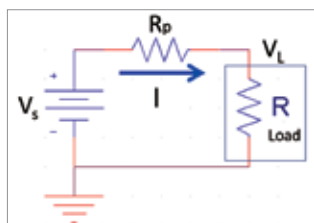


Fig. 1: Basic electrical model of a power delivery network

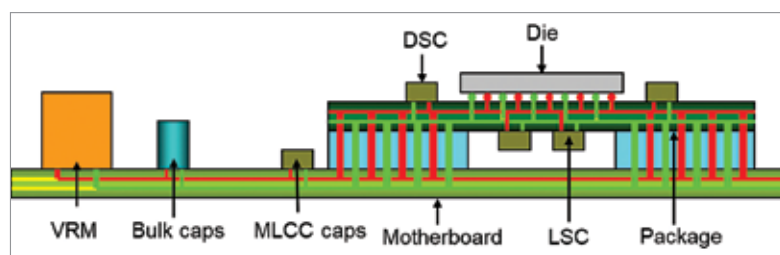


Fig. 2: Power delivery network of a CPU on desktop or server motherboard

Advantages for the designer

Power integrity plays a significant role in the success and failure of electronic products. Following are the key advantages of performing thorough power integrity analysis of a system:

- Ensures design specifications are met at all system level components during worst-case operating conditions.
- Ensures the system is designed optimally, with right level of inbuilt safety margins, making it cost-effective.
- Stronger reliance on results based decisions rather than experience or thumb-rule based design.
- Allows system scalability and optimisations based on individual use cases. Need not rely on vendor guidelines based on generic designs that may not be cost-effective for a dedicated application.

Power delivery networks modelling

All electrical products, at basic level, can be reduced to electrical networks for modelling purpose. A power delivery network (PDN) comprises majorly a power source, single or multiple loads, and an interconnect path between the source and the load.

Fig. 1 shows a simplistic electric circuit equivalent of a PDN with load R connected to voltage source V_s through path resistance R_p . In a real system, such as a desktop or server motherboard, V_s will be a voltage regulator (VR) powering up the CPU, which is modelled as R (load). The parasitic resistance of the layout/path on board connecting VR and